

Do Function Vectors and Task Vectors Generalize?

A Reproducibility Study Across Model Families and Scale

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Paper, code and data
serendeeep.tech/research/
function-task-vectors
github.com/Serendeeep/fv-tv-repro

Summary: where each vector beats its control

On word pairs, mostly yes, except our Gemma FVs and the held-out Llama-8B TV. On sentiment, no. TVs stay near chance with arbitrary or natural labels.

	Test 1 Exploratory grid	Test 2 Held-out layers	Test 3 Sentiment
method, model	5–16 tasks. Layer picked on the scored data. Share of ICL gap recovered.	8 tasks. Vectors and controls each pick a layer on dev data. Points above control.	SST-2 reviews, A/B labels. TV accuracy ; chance is 50%. TV minus shuffled: all CIs include 0.
FV GPT-J-6B	✔ 0.50 control −0.01	✔ +35.9 CI 18.5 to 52.6	⊖ not run
TV GPT-J-6B	✔ 0.56 shuffled 0.28, $d = 0.78$	✔ +16.1 CI 4.1 to 30.3	⊖ 52.7% ICL 73.5%
FV Llama-3.1-8B	✔ 0.42 control 0.00	✔ +24.3 CI 7.6 to 42.9	⊖ not run
TV Llama-3.1-8B	✔ 0.83 shuffled 0.42	⊖ −4.2 CI −20.7 to 8.7	⊖ 50.0% ICL 90.8%
TV Llama-3.1-70B	✔ 0.80 shuffled 0.48	⊖ not run	⊖ not run
FV Gemma-2-9b base	⊖ not run	✖ 0% acc. 2 tasks, ICL 70–99%	⊖ not run
TV Gemma-2-9b base	⊖ not run	✔ +19.1 CI 3.9 to 35.3	⊖ 49.6% ICL 94.0%
FV Gemma-2-9b-it	✖ 0.01 control 0.01	✖ 0% acc. 2 tasks, ICL 68–91%	⊖ not run
TV Gemma-2-9b-it	✔ 0.83 shuffled 0.39	✔ +32.3 CI 7.4 to 57.9	⊖ 50.0% ICL 93.3%

Table 1: Tests 1→2 tighten layer selection; Test 3 changes the task. ✔ pass (Test 1: CI above 0, $d \geq 0.8$) or CI above 0 (Test 2, ungraded) ✔ weak pass ($0.5 \leq d < 0.8$) ⊖ CI includes 0 ⊖ near chance ✖ fail (Test 1) or 0% accuracy (Test 2)
⊖ not run Gemma cells in Tests 2 and 3 ran on Kaggle; all other runs used NDIF.

Background

Function vectors (FVs)^[1] add the mean outputs of a few causally chosen attention heads (by AIE) to the residual stream. Task vectors (TVs, θ)^[2] copy one hidden state from a demonstration prompt into a zero-shot pass. Both papers say the vector carries the task.

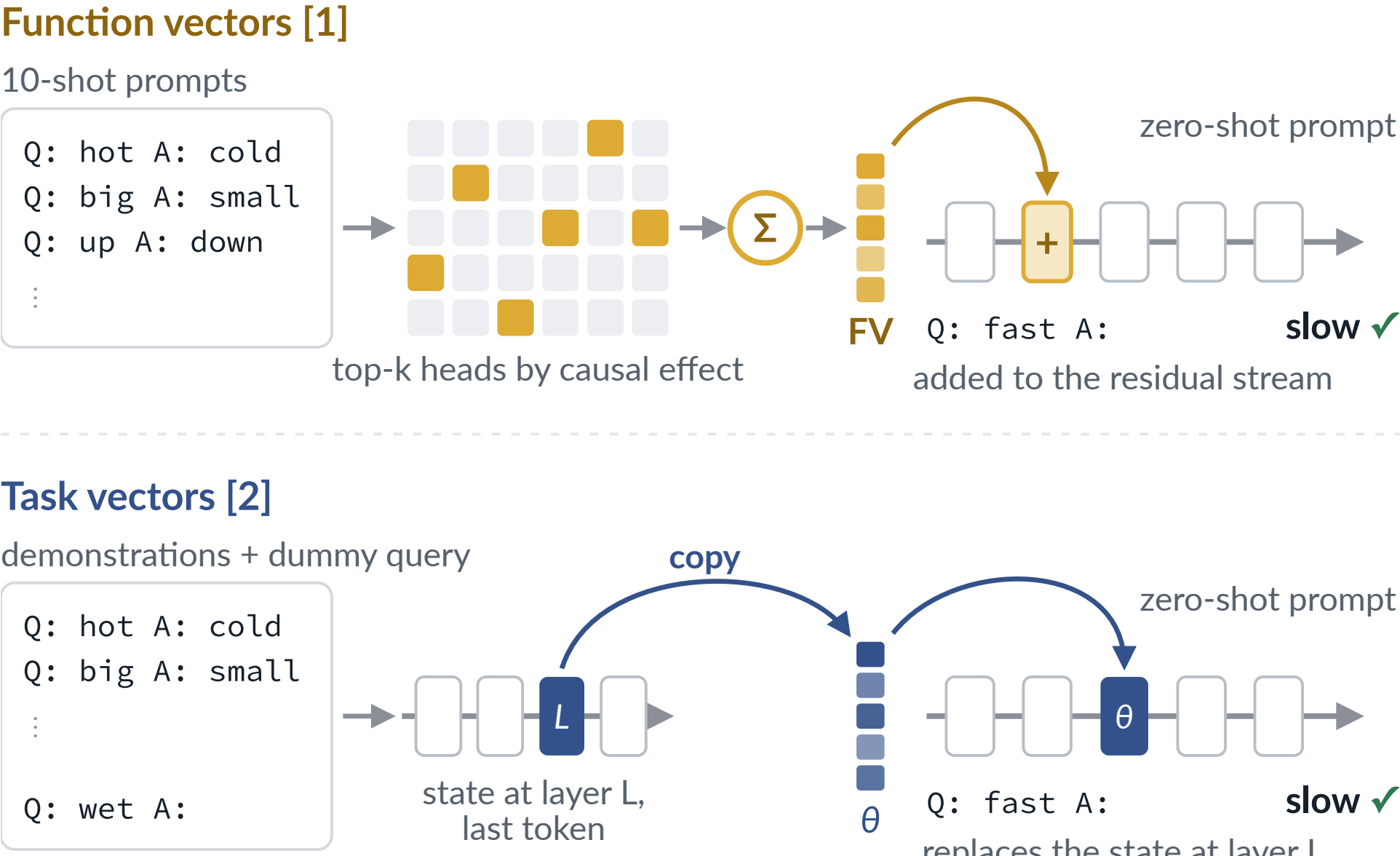


Figure 1: The two methods, reimplemented in NNSight^[3].

Setup

- GPT-J-6B, Llama-3.1-8B and 70B, Gemma-2-9b base and IT.
- 16 word-pair tasks from [1]; first-token accuracy.
- Random vectors and heads control FVs; shuffled labels control TVs.

Test 1: Exploratory grid

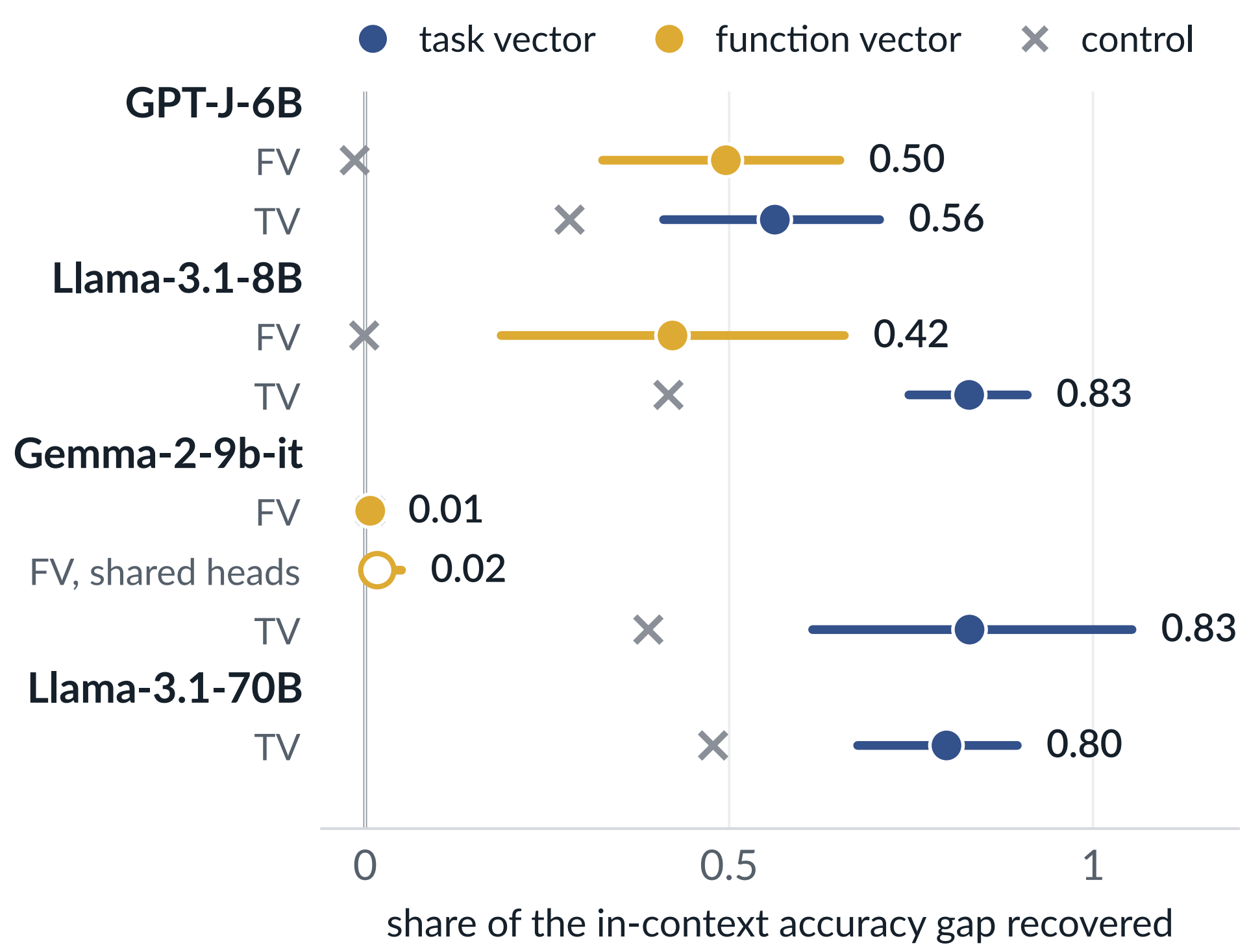


Figure 2: Share of the ICL gap recovered, with 95% CIs. x: control. Hollow: shared heads ($k = 40$). Picking the layer on scored data favours the real vector.

Scaling the Gemma FV up to 8×, moving it before the RMSNorm, or using 40 shared heads (closer to [1]) does not rescue it.

Test 2: Held-out layer selection

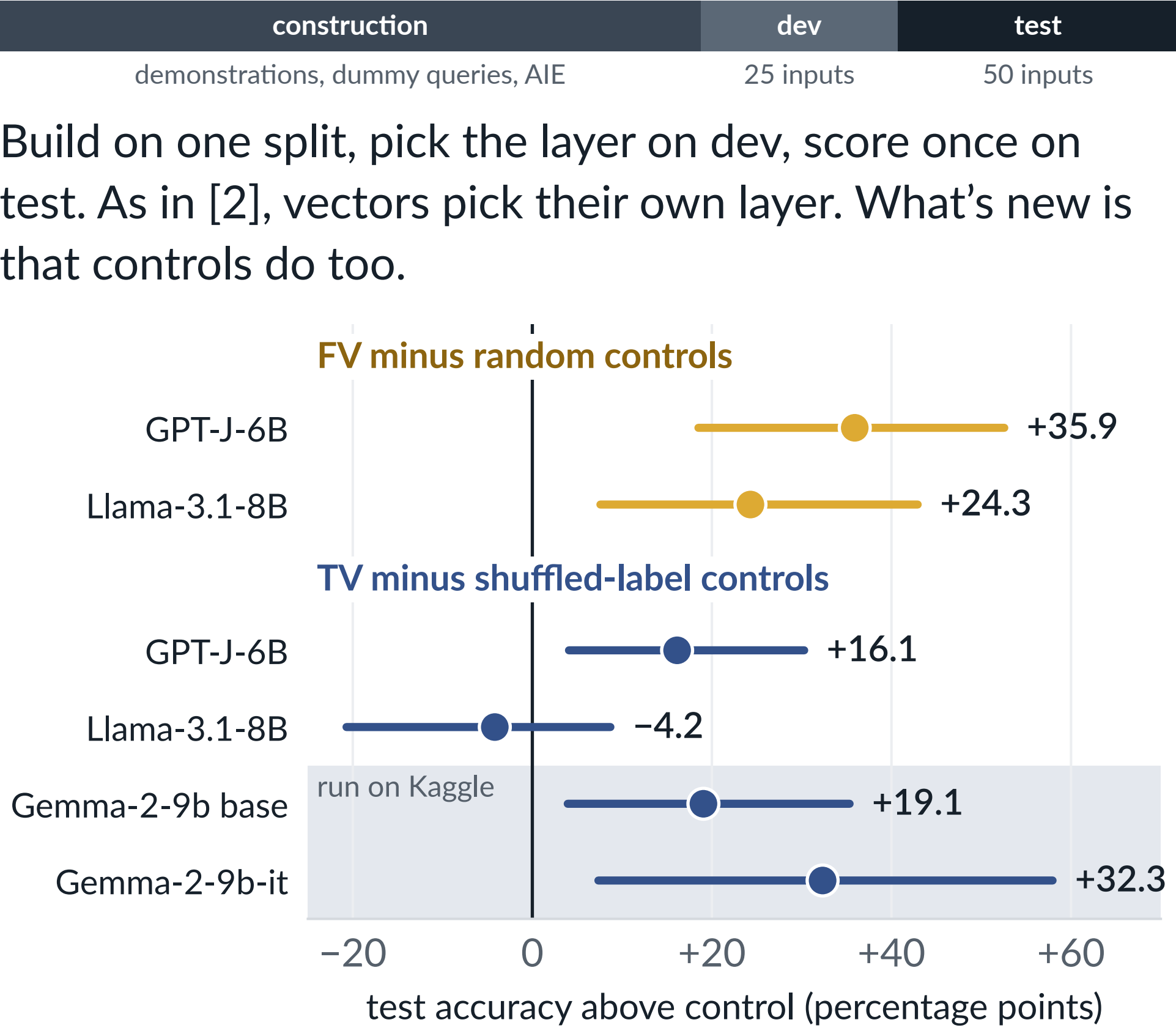


Figure 3: Points above control, 95% CIs over 8 tasks.

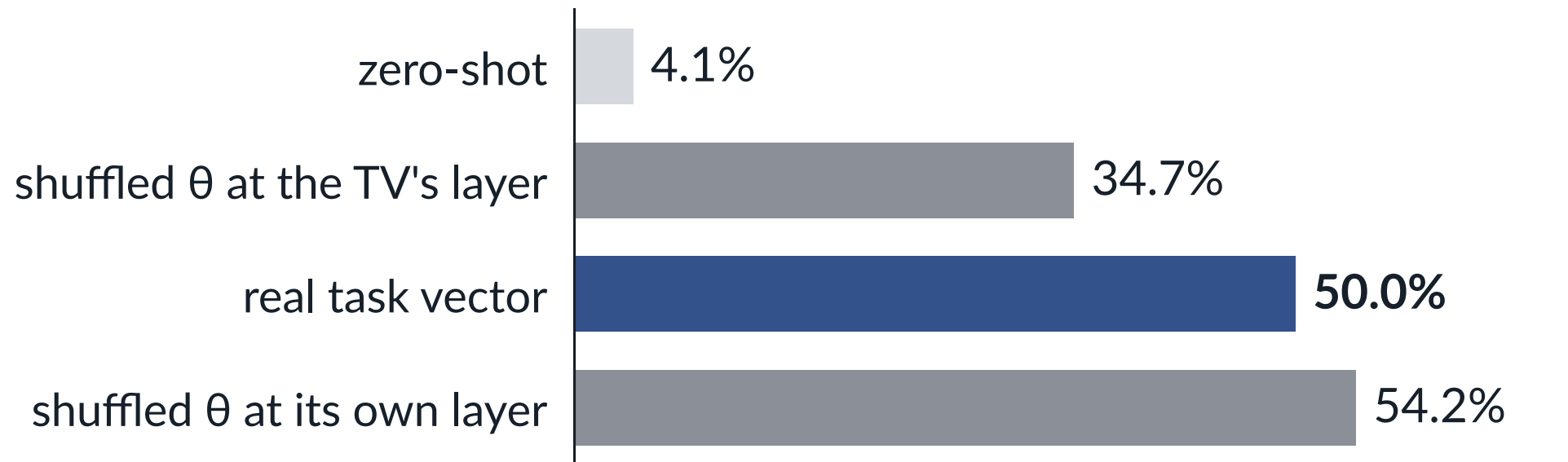


Figure 4: Llama-3.1-8B. Given its own layer, the shuffled control beats the real TV.

The gap across tasks is −4.2 points (CI −20.7 to 8.7), so the result is inconclusive.

Gemma: base vs instruction-tuned

Same splits and demonstrations for both. Kaggle matches NDIF on 197 of 200 checked predictions.

FV, test accuracy (%)	ICL	FV	control
base, antonym	70.0	0.0	0.0
base, country-capital	98.7	0.0	0.0
IT, antonym	68.0	0.0	0.0
IT, country-capital	90.7	0.0	0.0

The base model fails too, so instruction tuning alone doesn't explain it.

Limitations

- Our recipes differ from [1] and [2] (FV: 10 heads per task, no ICL-correct filter; TV: Q:/A: prompts). The Gemma null is about our implementation.
- Tests 2 and 3 use 6B–9B models only. Test 3 is TV-only.
- ICIs resample 8 tasks; test inputs are fixed.
- Follow-ups were chosen after earlier results.

Test 3: Sentiment with arbitrary labels

Positive maps to A and negative to B, or the reverse, so label meaning can't help^[4]. ICL learns this. The TV doesn't.

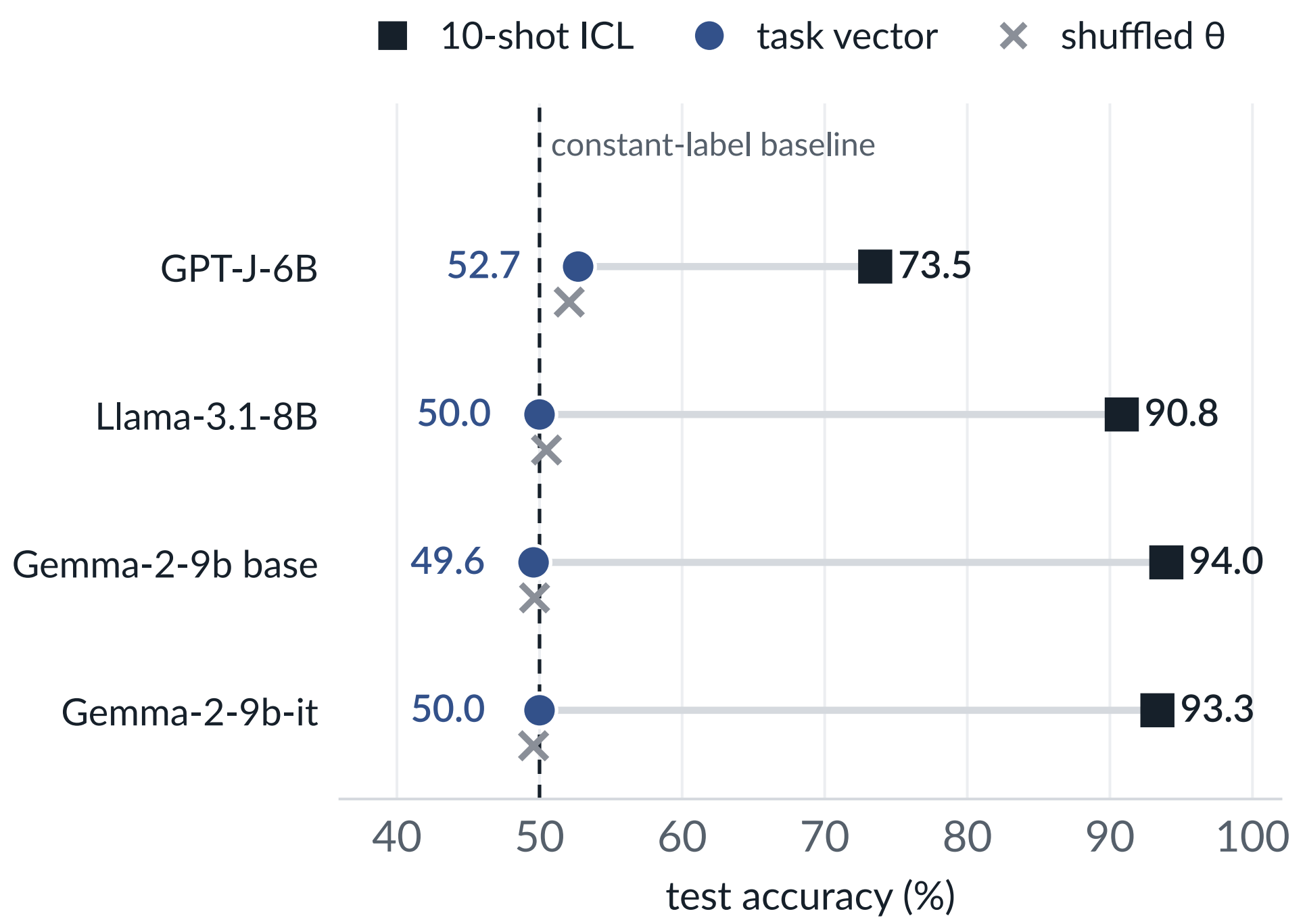


Figure 5: Mean of both mappings, 3 seeds. x: shuffled control.

In 18 of 24 cases the patched model gives every review the same answer.

Shuffled theta still works

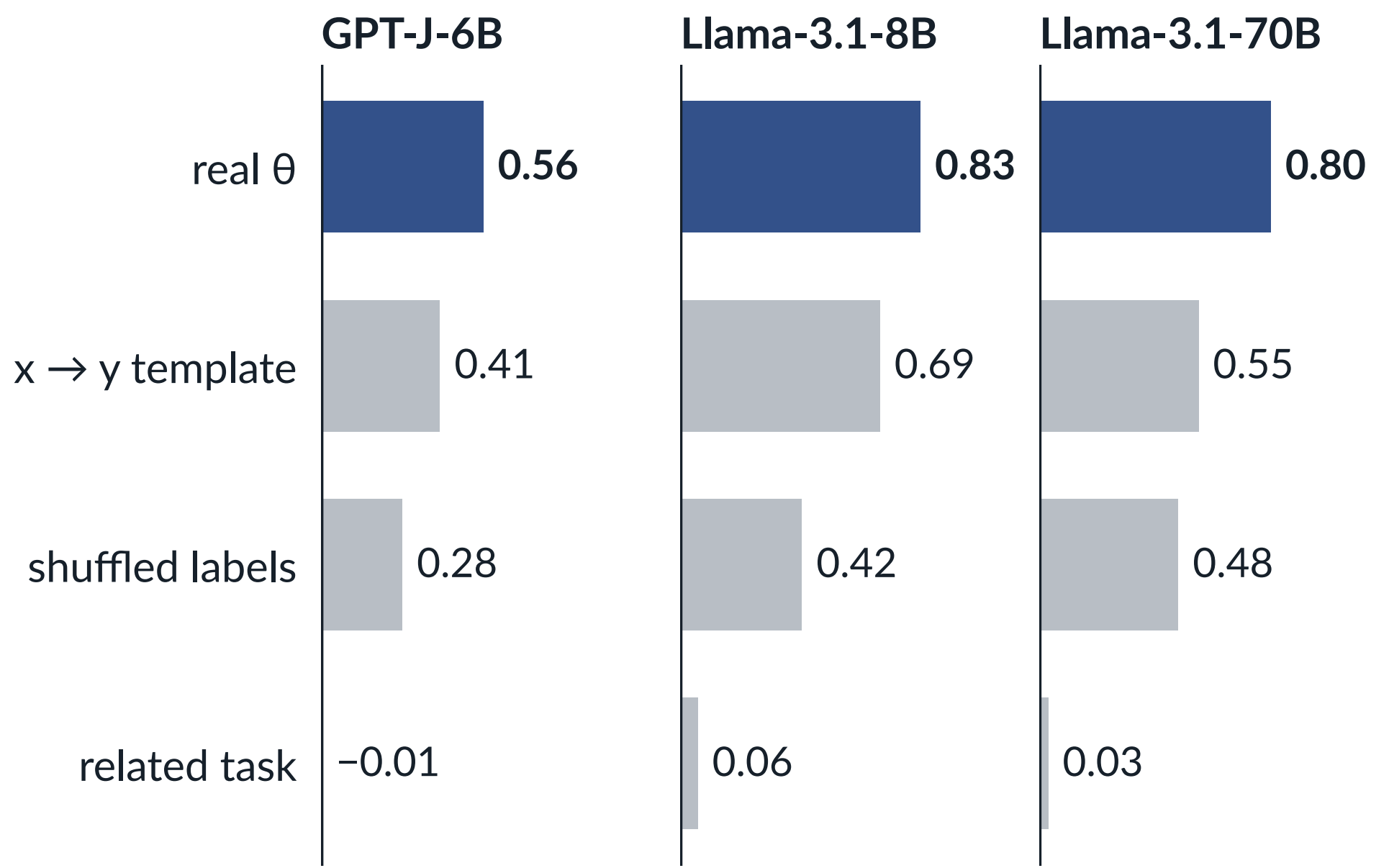


Figure 6: Test 1 recovery at the real TV's layer.

Shuffled labels keep about half the effect, consistent with task recognition^[5,6]. A related task's θ does almost nothing.

Conclusions

- Word-pair success does not imply broader transfer. On sentiment, TVs stay near chance on all four checkpoints.
- Controls need their own held-out layer. On Llama-8B, it removes the TV's apparent win.
- Our FV works on GPT-J and Llama-8B but not on Gemma. The cause is open.

References. [1] Todd, Li, Sharma, Mueller, Wallace, Bau. Function vectors in large language models. ICLR 2024. [2] Hendel, Geva, Globerson. In-context learning creates task vectors. Findings of EMNLP 2023. [3] Fiotto-Kaufman et al. NNSight and NDIF: Democratizing access to open-weight foundation model internals. ICLR 2025. [4] Wei et al. Larger language models do in-context learning differently. arXiv:2303.03846, 2023. [5] Min et al. Rethinking the role of demonstrations: What makes in-context learning work? EMNLP 2022. [6] Pan, Gao, Chen, Chen. What in-context learning "learns" in-context: Disentangling task recognition and task learning. Findings of ACL 2023.

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